Theory section (skeleton): consistency of the reported model, evidence and inclusion probabilities for SMC-BVS

Draft for the JRSS-B paper — S-PAPER, branch `theory/S-PAPER`

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Reading guide. Every numbered result below is a candidate statement assembled from the verification campaign. The comment lines in the LaTeX source give, for every statement, its author status, the branches that verified it and the exact source label. In this typeset text, the phrase "Proof: see Supplement § X (source branch ...)" stands for a proof that is *not* reproduced here. Where the source is a single-run proof not checked by a second run, this is said in a remark, and the status words of the campaign are never upgraded.

1 Setting and notation

1.1 Data, target and decision objects

Let (X^r, y^r) be raw data, $n \geq 3$, with X^r having no constant column and y^r non-constant. With $C = I - \mathbf{1}\mathbf{1}^\top/n$ and D_x the diagonal matrix of retained raw column standard deviations, the standardised data are $X = CX^r D_x^{-1}$ and $y = Cy^r/s_y$ with $s_y^2 = \|Cy^r\|^2/(n-1)$, so that $\|x_j\|^2 = \|y\|^2 = n-1$. The state space is $\Omega = \{S \subseteq [p] : |S| \leq K\}$ with $K = \min(p, \lfloor n/2 \rfloor)$. For $S \in \Omega$ write $G_S = X_S^\top X_S = (n-1)R_S$, $z_S = X_S^\top y$, $H_S = (n+1)G_S + I$, $D_S = \det H_S / \det(G_S + I)$, $Q_S = y^\top y - n\, z_S^\top H_S^{-1} z_S$ and $\nu = 0.01 + n/2$. The target is the Bayesian variable-selection posterior with $\beta_S \sim N(0, n\sigma^2(G_S+I)^{-1})$, $\sigma^2 \sim \mathrm{IG}(0.01, 0.01)$ and a beta–binomial size prior $\mathrm{BB}(1, p)$ capped at K: the marginal likelihood is $m(S) = (2b_0)^{a_0}\Gamma(\nu)\pi^{-n/2}\Gamma(a_0)^{-1} D_S^{-1/2}(2b_0 + Q_S)^{-\nu}$ ($a_0 = b_0 = 0.01$), the prior is $\varpi(S) = w(|S|)/W_K$ with $w(k) = \frac{1}{2}\binom{2p-1}{k}^{-1}$ and $W_K = \sum_{k \leq K}\binom{p}{k} w(k)$, and

$$F(S) = \log[\varpi(S)m(S)], \qquad Z = \sum_{S \in \Omega} \varpi(S)m(S), \qquad \pi(S) = e^{F(S)}/Z.$$

Write $\mathrm{PIP}_j = \pi(j \in S)$ and $\mathrm{MPM} = \{j : \mathrm{PIP}_j > \frac{1}{2}\}$. The *global* HPM is $\arg\max_\Omega F$ with ties broken by smaller size, then lexicographically. The *reported* HPM $\widehat{S}$ is the same rule applied to the set $\mathcal{S}$ of all models ever scored.

1.2 Truth and asymptotic regime

The statistical model is $y^r = \alpha^* \mathbf{1} + X^r\beta^* + \varepsilon$ with $\mu = CX^r\beta^* = X\theta^*$, $\theta^* = D_x\beta^*$, $S_0 = \operatorname{supp}\beta^*$ and $s_0 = |S_0|$; P_0 is the law of ε and "joint probability" adds the algorithm's randomness. Put $\delta_1 = 1 - \pi(S_0)$ and $r_n = \sqrt{\log n/n}$. **Regime R1** is the triangular array $n \to \infty$, s_0 fixed, $p = p_n \leq n^A$ for a fixed $A \geq 1$. Two design/model bundles are used.

- **G**: fixed design; $\varepsilon \sim N_n(0, \sigma_*^2 I)$ with $\sigma_*^2 > 0$ fixed; intercept removed by contrasts.
- **R**: s_0 fixed; $p \leq n^A$ with $A \geq 1$; $\|\mu\|^2/n \leq M$; $\lambda_{\min}(R_{S_0}) \geq \varphi_0^2 > 0$; $s_0 \leq K$ eventually.

1.3 The sampler, in the notation the results use

The method partitions $[p]$ into blocks $C_1, \dots, C_J$ (deterministic in X; complete linkage, top-down cut, next-fit packing) of size at most $B = \min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, 16)$, and computes block tables $r_b(A) = \mathrm{BF}(A) = m(A)/m(\emptyset)$ for $A \subseteq C_b$ against the full response. The *composite start* is $\pi_0 = \gamma_0/\bar{Z}_0$ with $\gamma_0(S) = \varpi(S)\prod_b r_b(S \cap C_b)$, sampled exactly; the *start mode* is $\tilde{S}_0 = \arg\max \pi_0$ (computed exactly by max-plus convolution of the block tables). A tempering bridge $\gamma_\phi = \gamma_0 e^{\phi d}$, $\phi \in [0,1]$, with $d(S) = \log \mathrm{BF}(S) - \sum_b \log r_b(S \cap C_b)$, connects $\pi_0 = q_0$ to $\pi = q_1$. One round at level ϕ consists of p Gibbs flips, one uniform swap, B neighbour swaps and one independence proposal from π_0, in a uniformly random order. The calibration $\mathcal{C}$ (grid, number of stages T, burn-in $L = \max(4, \lceil\log_2(n+1)\rceil)$, population size N_e, tier, ...) is a function of the data and calibration streams only. The final run uses 4 islands of N_e exact i.i.d. draws from π_0, the frozen grid, L burn-in rounds and 4 measured rounds at $\phi = 1$, and a local search of at most $\ell = \lceil\log_2(n+p+1)\rceil$ moves whose scans score every feasible flip and swap of the (frozen) centre. Outputs are $\widehat{Z}_r$ (island evidence), $\widehat{Z} = \frac{1}{4}\sum_r \widehat{Z}_r$, the bound $0.05\widehat{Z}$, the Rao–Blackwell islandwise PIPs $\widehat{\mathrm{PIP}}_{rj}$, the primary PIP $\bar{P}_j = \frac{1}{4}\sum_r \widehat{\mathrm{PIP}}_{rj}$, $\widehat{\mathrm{MPM}}$ and $\widehat{S}$. The final run scores $Q_f := 4N_e(T + L + 5)$ i.i.d. draws from π_0, with $4N_e(L+6) \le Q_f$.

1.4 Runs, outputs, scored sets and LS0

Definition 1 (Potential run and output events)**.** The *potential uninterrupted run* $\bar{R}$ is the record the method produces under exact arithmetic with the resource preflight and the in-run time guard removed. Let $\mathcal{G} = \{\text{status} = \texttt{resource_limit}\}$ and let $U \supseteq \mathcal{G}$ be the no-output event (with potential-run part $\bar{U}$). Write $\mathcal{O} = U^c$ for the event that output is available (status `exact` or `completed`), $\mathcal{O}_E$ when evidence and PIPs are reported and $\mathcal{O}_H$ when an HPM is reported; $\bar{\mathcal{O}} := \bar{U}^c$ is the output-available event of the potential run. $\bar{\mathcal{S}}$ is the scored set of $\bar{R}$ (calibration, final run, rejected proposals, *both* Gibbs configurations at each flip, local-search scans).

Definition 2 (LS0: the final local search from both the best-scored model and the start mode)**.** The method's final local search is the production local search (at most ℓ moves, all feasible flips and swaps scanned, strict-improvement moves, lexicographic tie rule) centred at the running HPM. **LS0** is the one extra production local search centred at the start mode $\tilde{S}_0 = \arg\max \pi_0$, with its centre emitted as a score event *before* the search begins and with the same HPM tie rule. Let $\mathcal{S}_{\mathrm{LS0}}$ be the models it scores and

$$\mathcal{S}^+ := \mathcal{S} \cup \mathcal{S}_{\mathrm{LS0}}, \qquad \widehat{S}^+ := \text{the best model of } \mathcal{S}^+ \text{ under the method's tie rule.}$$

Without LS0, $\mathcal{S}^+ = \mathcal{S}$ and $\widehat{S}^+ = \widehat{S}$. When LS0 is part of the method (the presentation approved by the project lead on 2 November 2026, conditional on the data check O-LS0, which was not available when this draft was written), the reported HPM is $\widehat{S}^+$, and $\bar{\mathcal{S}}^+$ denotes the corresponding potential scored set.

1.5 Notation for the repair theory

For a model S put $\Phi(S) = |S \triangle S_0|$, $m(S) = |S_0 \setminus S|$ (missing truths) and $e(S) = |S \setminus S_0|$ (extras), so $\Phi = m + e$. A flip or swap neighbour T of S is a *progress* neighbour if $\Phi(T) < \Phi(S)$: adding a missing truth, deleting an extra, or swapping an extra for a truth. For $S = U \cup P$ with $U \subseteq S_0$, $P \cap S_0 = \emptyset$, the lattice is $\mathrm{Lat}(S) = \{U' \cup P' : U \subseteq U' \subseteq S_0,\ P' \subseteq P\}$, of size $2^{\Phi(S)}$. $\mathcal{R}_{a,d} = \{S = U \cup P : U \subseteq S_0,\ m(S) \le a,\ P \cap S_0 = \emptyset,\ |P| \le d\}$ (feasible, $|S| \le K$), and $\mathcal{R}_d = \mathcal{R}_{s_0,d}$. $M(S)$ is the production best-move map ($\perp$ if no strictly improving feasible flip or swap). D_S is the event that

the best progress neighbour of S strictly beats both S and every non-progress neighbour. P_T is the orthogonal projection on span X_T and $\Delta(T) = \|(I - P_T)\mu\|^2/(n-1)$ is the population residual (deterministic under G).

2 Standing assumptions

The following five declared assumptions are used throughout; each theorem lists any further hypotheses itself.

Assumption 1 (EX-A)**.** Exact scores, convolutions, probabilities and comparisons (exact arithmetic).

Assumption 2 (EX-S)**.** Independent ideal uniform streams with the method's stream separation.

Assumption 3 (CAL)**.** The calibration $\mathcal{C}$ uses streams 0–9 only; the final streams 10–13 are fresh; escalation is decided before any final randomness. In particular the calibration is a function of the data and of streams 0–9 only, and every fixed-data identity below conditions on a successful frozen $\mathcal{C}$, never on later diagnostics.

Assumption 4 (SR)**.** Each Gibbs update evaluates both feasible configurations (the current state S and the flipped state $S\triangle\{j\}$), and both count as scored: "scored" means offered to the HPM store at least once in the fit. An infeasible addition ($|S| = K$, $j \notin S$) is not evaluated.

Assumption 5 (GUARD)**.** The resource guard is a stop-only coupling to $\bar{R}$: on $\mathcal{G}^c$ the status and every reported output equal those of $\bar{R}$. No assumption is made on the law of $\mathcal{G}$ given $\bar{R}$. (Operationally: no guard path touches a random-number generator or a setting.)

Remark 1 (Two further operational assumptions). **START**: the start draws are exact i.i.d. π_0 draws with exact $\bar{Z}_0$. **OUT** (used by Theorem 6): on $\mathcal{O} \cap \{$SMC branch taken, calibration succeeded$\}$ the reported outputs equal the potential run's; on $\mathcal{O} \cap \{$exact enumeration branch$\}$ they are exact. In all statements below, *nothing is ever conditioned on* $\mathcal{O}$: conclusions have the form $\mathrm{P}(\mathcal{O} \cap \{\text{error}\})$.

3 The story

SMC-BVS reports a consistent model because the truth gets *scored* and the posterior then *ranks* it first. The reported HPM is the best model ever scored: every table and scan, both Gibbs configurations of every flip, rejected proposals, and (with LS0) the local search from the start mode. Consequently the error decomposes, for every n and every data set, as

$$\mathrm{P}(\text{output and } \widehat{S}^{+} \neq S_0) \;\le\; 2\,\mathrm{E}_0[1 - \pi(S_0)] \;+\; \mathrm{P}(\text{the output-potential run never scores } S_0) \quad (1)$$

(Theorem 4). The paper proves that the two terms vanish, in two different ways. The *posterior* term vanishes by Theorem 1 (selection consistency under anchored, size-restricted sparse-eigenvalue conditions and beta-min). The *discovery* term vanishes in one of two regimes.

- **Favourable regime.** The start already finds the truth: $1 - \pi_0(S_0) \to 0$ (Theorem 2(iii)), so the final run's i.i.d. π_0 draws score S_0 (Theorem 5(a)).
- **Masked regime.** The start mass at the truth vanishes, $\pi_0(S_0) \to 0$ (variables whose contribution cancels across blocks cannot be seen by any block table), but the start concentrates *near* the truth at a reference model S° (Theorem 2(iv)), and a deterministic greedy repair from the reference scores the truth (Theorem 3(i), repair), which LS0 delivers from the start mode (Theorem 5(b1)).

Theorem 6 says that, in the favourable regime, the *other* reported quantities (evidence, inclusion probabilities) are consistent too; Proposition 2 says that the frozen-run identities and the 95% evidence bound hold at every n for the potential run, and at the output level under GUARD; Proposition 1 says that the HPM is the right decision under 0–1 loss and names the one within-size error that no size-prior retuning can fix. What the story cannot say is anything numerical at $n = 200$: this is the subject of Section 5, and it must be read with every theorem.

4 Results

4.1 Theorem 1: selection consistency (T03$^\sharp$) and the exact size tail

Notation. Let $q_k = \binom{p}{k} w(k)$ and

$$R_m \;=\; \frac{\varpi(\{S \in \Omega : |S| > m\})}{\varpi(S_0)} \;=\; \sum_{k=m+1}^{K} \frac{q_k}{w(s_0)}$$

(a data-free number; $R_m = 0$ for $m \geq K$). For real ω let $m_n^\sharp(\omega) = \min\{m \geq s_0 : \; R_m \leq n^{-(s_0+1)/2} e^{-\omega}\}$ and $m_n^\sharp := m_n^\sharp(\frac{1}{2}\log n)$ (so $s_0 \leq m_n^\sharp \leq K$). Let $j_0 = \lceil (2A+1)s_0 \rceil + 1$ and fix $g_0 > 0$.

Definition 3 (Hypotheses of Theorem 1)**.**

- $\mathrm{SE}_\lambda(j_0)$ *(anchored, normalised, bounded size)*: $\lambda_{\min}(R_{S_0 \cup J}) \geq \varphi_0^2$ for every $J \subseteq [p] \setminus S_0$ with $|J| \leq j_0$ and $S_0 \cup J \neq \emptyset$.
- $\mathrm{SE}_w(m; g_0)$ *(anchored, unnormalised)*: $\lambda_{\min}(G_{S_0 \cup J}) \geq g_0$ (equivalently $\lambda_{\min}(R_{S_0 \cup J}) \geq g_0/(n-1)$) for every $J \subseteq [p] \setminus S_0$ with $|J| \leq m$ and $S_0 \cup J \neq \emptyset$.
- BM-∞: $\min_{j \in S_0} \theta_j^{*2} \geq M_n \sigma_*^2 \log n/(n\varphi_0^2)$ with $M_n \to \infty$. BM(c): the same with a fixed constant $c > c_0(s_0, A)$, where $c_0(s_0, A) < \infty$ depends only on s_0 and A (the proof gives $c_0 = L_* = \max\{4C_*, 32[s_0(A+1)+2]\}$; this is a proof artefact, not sharp).

By eigenvalue interlacing, "anchored" is equivalent to the unanchored condition on every column set T with at most j_0 (resp. m) columns outside S_0; in particular $\mathrm{SE}_w(m; g_0)$ requires every m columns of X to be linearly independent, with $\lambda_{\min}(X^\top X) \geq g_0$ on them.

Lemma 1 (Exact size tail, T02(i$^\sharp$))**.** *Assume G and R, and let n be such that $s_0 \leq K$. For every integer m define $\Theta_{n,m}^\sharp \geq 0$ by $\Pi(|S| > m \mid y) = R_m\,(n+1)^{s_0/2}\sqrt{n}\;\Theta_{n,m}^\sharp$ if $R_m > 0$ and $\Theta_{n,m}^\sharp := 0$ if $R_m = 0$. Then for every $L > c^{\sharp 2}$ (with $c^\sharp$ and C_Λ explicit constants depending on σ_*^2, M, φ_0 only)*

$$\sup_m \mathrm{P}_0(\Theta_{n,m}^\sharp > L) \;\leq\; \frac{8 + 24M/\sigma_*^2}{n-1} + \frac{1}{\sqrt{L}} + \frac{C_\Lambda}{\log(\sqrt{L}/c^\sharp)}.$$

Consequently, for every deterministic real sequence ω_n, $\Pi(|S| > m_n^\sharp(\omega_n) \mid y) = O_{\mathrm{P}_0}(e^{-\omega_n})$; in particular $\Pi(|S| > m_n^\sharp \mid y) = O_{\mathrm{P}_0}(n^{-1/2})$, with $m_n^\sharp \leq \min\{K, m_n^{\mathrm{old}}\}$. The inequality is non-asymptotic. No sparse-eigenvalue condition is used.

Proof: see Supplement §S.1 (source branch `theory/W1-T03R`, `PROOF.md` §§1–2).

Lemma 2 (Superset mass, T02(ii$^\sharp$))**.** *In regime R1 under G, R and $\mathrm{SE}_w(m_n^\sharp; g_0)$: $\Pi(\{S \in \Omega : S \supsetneq S_0\} \mid y) = O_{\mathrm{P}_0}(r_n)$.*

Proof: see Supplement §S.1 (source branch `theory/W1-T03R`, `PROOF.md` §5).

Theorem 1 (Selection consistency, T03$^\sharp$)**.** *Assume regime R1, G, R,* $\mathrm{SE}_\lambda(j_0)$*,* $\mathrm{SE}_w(m_n^\sharp; g_0)$ *for a fixed* $g_0 > 0$*, and a beta-min condition in one of the forms* BM-∞ *or* BM(c) *of Definition 3 (for (i)–(iii)). No hypothesis H-T02 is used (Theorem 1 is used directly). Then:*

(i) $1 - \pi(S_0) = O_{\mathrm{P}_0}(r_n)$*;*

(ii) $\max_j |\mathrm{PIP}_j - \mathbf{1}\{j \in S_0\}| \le 1 - \pi(S_0)$ *for* every *data set, and* $\mathrm{P}_0(\mathrm{MPM} = S_0) \to 1$*;*

(iii) $\mathrm{P}_0(S^\dagger \neq S_0) \le 2\mathrm{E}_0[1 - \pi(S_0)] \to 0$*, where* $S^\dagger$ *is the global HPM; on* $\{\pi(S_0) > \frac{1}{2}\}$*,* S_0 *is the strict global maximiser of* F *(so* $\mathrm{HPM} = \mathrm{MPM} = S_0$ *strictly);*

(iv) (no beta-min) for fixed $c > c_0$ *and* $J_n(c) = \{j \in S_0 : \theta_j^{*2} \ge c\,\sigma_*^2 \log n/(n\varphi_0^2)\}$*:* $\Pi(S \not\supseteq J_n(c) \mid y) = O_{\mathrm{P}_0}(r_n)$ *and* $1 - \min_{j \in J_n(c)} \mathrm{PIP}_j = O_{\mathrm{P}_0}(r_n)$*; the same holds for* $J_n^\infty = \{j : \theta_j^{*2} \ge M_n \sigma_*^2 \log n/(n\varphi_0^2)\}$ *with any* $M_n \to \infty$*;*

(v) scale invariance of the conclusions.

Proof: see Supplement §S.1 (source branch `theory/W1-T03R`, `PROOF.md` §6; locked T01–T03 as cited there).

Remark 2 (Scope of Theorem 1).

1. The "anchoring" of the eigenvalue condition is cosmetic (interlacing; Definition 3). The real relaxations relative to the locked SE are the size range ($m_n^\sharp$ instead of m_n) and the normalisation (g_0 instead of $(n-1)\varphi_0^2$) on large unions.
2. The statement "the exact tail cuts m_n to $s_0 + O(s_0 \log n/\log p)$" is *false* under G and R alone (duplicated columns, W1-T03R Prop. D, COUNTEREXAMPLE, reproduced by W2-VT03R and A-REV2); the exact tail improves the locked size cutoff by a constant ($4s_0!\,e^{\gamma_m}$) and an $m^2/(4p)$ term, not by a power of p. [Status: PROVED/VERIFIED, W1-T03R E1(d), Prop. D.]
3. (PROOF SKETCH, W2-VT03R Prop. W.) $\mathrm{SE}_\lambda(j_0)$ + BM may be replaced by SE_w plus a direct signal-separation condition SIG(c), $c \ge L_*$, for conclusions (i)–(iii).

4.2 Proposition 1: Bayes decisions and the size-prior ceiling (T07, with T06(i))

Proposition 1 (Decision rules and the size-prior ceiling)**.** *Let* π *be the target posterior and* S_k^* *the tie-order-least maximiser of the marginal likelihood among models of size* k*.*

(i) The HPM is a Bayes rule for 0–1 *loss on supports; the median probability model is a Bayes rule for Hamming loss.*

(ii) For any PIP estimate $\hat{p}$ *the plug-in Hamming rule has excess posterior risk at most* $2\|\hat{p} - \mathrm{PIP}\|_1$*.*

(iii) If $s_0 \le K$*, some positive size-only prior has global HPM* S_0 *if and only if* $S_{s_0}^* = S_0$*. In particular, if a scored model* $S' \neq S_0$ *with* $|S'| = s_0$ *has a higher score than* S_0*, no positive size-only prior recovers* S_0 *on those data.*

(iv) Two supports have equal score for every response if and only if $X_S X_S^\top = X_{S'} X_{S'}^\top$ *(outer products, not projections); they then have equal size.*

Proof: see Supplement §S.2 (source branch `theory/W2-BOOK`, `PROOF.md` §3; locked T07, T06).

Remark 3. Part (iii) concerns *within-size* errors ($|S| = s_0$, $m(S) \ge m(S_0)$) only; size-prior retuning *can* move different-size (superset or subset) errors. It covers every truncated BB(a, b) prior. For (iv), with the cap K the MPM keeps the $\min(K, N_+)$ largest PIPs above 1/2 and is unique iff no $\mathrm{PIP}_j = \frac{1}{2}$.

4.3 Theorem 2: the start — favourable and masked regimes (T22, T27)

Setting. G, R-lite(A, M) (meaning: $p \le n^A$ and $\|\mu\|^2 \le Mn$), the v3 partition, blocks of size at most 16, $\nu = 0.01 + n/2$. **SE-blk**(φ_0): $\min_b \lambda_{\min}(R_{C_b}) \ge \varphi_0^2$ (checkable on the realised blocks; by interlacing equivalent to $\lambda_{\min}(X_U^\top X_U/(n-1)) \ge \varphi_0^2$ for every nonempty U inside a block). **BO**: $X_{C_b}^\top X_{C_{b'}} = 0$ for $b \ne b'$. For a reference S° and block C_b, $U_b^\circ = S^\circ \cap C_b$, $\lambda_b(S^\circ) = \|(P_{C_b} - P_{U_b^\circ})\mu\|^2$ and $g_b(S^\circ) = \min_{U \subseteq C_b, U_b^\circ \not\subseteq U} \|(P_{U \cup U_b^\circ} - P_U)\mu\|^2$. The conditions VIS, NL_η, $\mathrm{NL}_{\mathrm{avg}}(\kappa)$, EXP′ are those of Section 3 of the common-definitions file. Specific conditions used below: **ZL**(S°) (exact zero leakage): $\lambda_b(S^\circ) = 0$ for all blocks; **SPREAD**(Δ_0): $\min_b \|(I - P_{C_b})\mu\|^2/(n-1) \ge \Delta_0 > 0$; **VIS-lin**$(c_V)$: $\min_b g_b(S^\circ) \ge c_V n$ with fixed $c_V > 0$; **BM-blk**: $\min_{j \in S_0} \theta_j^{*2} \ge M_n(\sigma^2 + 2M)\log n/(\varphi_0^2(n-1))$ with $M_n \to \infty$; **CL**(L) (chain leakage): along a deterministic addition chain $S^{(0)} \subset \cdots \subset S^{(c)} = S_0$ with $S^{(i+1)} = S^{(i)} + l_i$, $\max_i |\delta_{T_i, l_i}| \le L\sigma$, where $T_i = S^{(i)} \cap C_{b(l_i)}$ and $\delta_{T,l} = u_{T,l}^\top \mu$ ($u_{T,l}$ the unit residual of x_l on X_T); **H3**: $\mathrm{P}_0(\pi_0(S^{(0)}) \ge c_0) \to 1$.

Theorem 2(i) (deterministic start bounds)**.** (i) *The composite start puts mass at least* $(1-q)^{s+1}/(1+e)^s$ *on any feasible reference model* S° *(*$S^\circ \ne S_0$ *allowed), with* q, e *the table constants of the locked T22(i); the odds from a reference to* S_0 *are exact; and the tempering bridge keeps* $\sup_{\phi \in [0,1]}[1 - q_\phi(S_0)] \le \max\{1 - \pi_0(S_0), 1 - \pi(S_0)\}$. *Hence, if* $S_0 \in \Omega$, $\Delta_n := 1 - \inf_\phi q_\phi(S_0) = \max\{1 - \pi_0(S_0),\, 1 - \pi(S_0)\} < 1$ *exactly.*

Proof: see Supplement §S.3 (locked T22; source branch `theory/W2-T26`, `PROOF.md` Lemma 1).

Theorem 2(ii) (EXP′ is a theorem)**.** *Assume G, R-lite*(A, M) *and SE-blk*(φ_0). *Put* $c_* = \sigma^2/(8(M + 2\sigma^2))$. *There is an explicit* κ_{EXP} *(depending on* φ_0, σ, M, n *only) and an event* E_n *with* $\mathrm{P}_0(E_n) \to 1$ *such that on* E_n, every *within-block addition* (b, T, l) *(*$T \subseteq C_b \setminus \{l\}$*) satisfies*

$$\log \frac{r_b(T + l)}{r_b(T)} = -\tfrac{1}{2}\log(n+1) - \nu \log(1 - x_{T,l}) + R_{n,b,T,l}, \qquad |R_{n,b,T,l}| \le \kappa_{\mathrm{EXP}},$$

where $x_{T,l} = (\mathrm{RSS}_T - \mathrm{RSS}_{T+l})/\mathrm{RSS}_T$. *In particular EXP′ holds in its strict form (*$\rho_n = \max|R| = O_{\mathrm{P}_0}(1)$*); at finite* n *a data-computable certificate replaces* κ_{EXP}. *Consequently EXP′ is removed from the hypothesis list of the favourable-start statement, and* NL_η *may be weakened to* NL_C *(W1-EXP Prop. 2.1, PROOF SKETCH).*

Proof: see Supplement §S.3 (source branch `theory/W1-EXP`, `PROOF.md` §1; `theory/W3-VEXP`).

Theorem 2(iii) (favourable start)**.** *For a feasible reference* S° *(which may be* S_0*, or* $S^\circ \ne S_0$*),* $1 - \pi_0(S^\circ) \to 0$ *in probability at the rate stated under any one of:*

(F1) the locked T22(iii) hypotheses minus EXP′: G, R, SE-blk, VIS, NL_η, $\mathrm{NL}_{\mathrm{avg}}(\kappa)$ *(all at reference* S°*): rate* $O_{\mathrm{P}_0}(2^B\sqrt{\log n}\, n^{-\kappa})$*;*

(F2) BO and BM-blk (with G, R, SE-blk, reference S_0*): rate* $O_{\mathrm{P}_0}(2^B\sqrt{\log n}\, n^{-1/2})$*;*

(F3) G, R-lite, SE-blk, ZL(S°)*, SPREAD, VIS-lin*(S°)*: rate* $O_{\mathrm{P}_0}(2^B n^{-1/2})$*, self-contained (no locked-proof transfer).*

Proof: see Supplement §S.3 (source branch `theory/W1-EXP`, `PROOF.md` §2, Theorem 6; verification `theory/W3-VEXP`).

Theorem 2(iv) (masked start)**.** *Assume G, R-lite, SE-chain (implied by SE-blk), a deterministic fixed-length addition chain, CL(L) with L fixed, H3 and $p \geq 2(s+c)$ ($s = |S^{(0)}|$, $c \geq 1$ the chain length). Then with probability $\to 1$, for all $i < c$,*

$$-1 - \kappa_{\mathrm{EXP}} \;\leq\; \log \frac{r_{b_i}(T_i + l_i)}{r_{b_i}(T_i)} + \tfrac{1}{2}\log n \;\leq\; \kappa_{\mathrm{EXP}} + 1.01(L + |Z_i|)^2 \quad (n \geq 400),$$

with $Z_i \sim N(0,1)$, and

$$\log \pi_0(S_0) + c \log p + \tfrac{c}{2}\log n = O_{\mathrm{P}_0}(1), \qquad \textit{i.e.} \quad \pi_0(S_0) = p^{-c} n^{-c/2} e^{O_P(1)}.$$

With $L = L_n \to \infty$ the upper side gains the term $1.01c(L_n + O_P(1))^2$ (a sufficient bound only). This is a statement about i.i.d. draws from π_0, not *about the hybrid sampler.*

Proof: see Supplement §S.3 (source branch `theory/W1-EXP`, `PROOF.md` §3).

Theorem 2(v) (one family carries both)**.** ***The family LH*** *(linked hubs): G groups of size g with equicorrelation $\rho_h \in [\rho_-, \rho_+] \subset (0,1)$, a hub in each group; groups independent except c disjoint* linked pairs *(a_i, b_i, ψ_i) with cross-covariance through the hubs; truth $\theta_{j_{a_i}} = \beta_i \neq 0$, $\theta_{j_{b_i}} = -\psi_i\beta_i$ (the* suppressor*), further unlinked visible groups with fixed nonzero coefficients; $s_0 = 2c + \#$visible fixed. The masked set is $M = \{j_{b_1}, \ldots, j_{b_c}\}$ and $S^\circ = S_0 \backslash M$.* Exact-Gram realisation: *$X^\top X = (n-1)\Sigma_{\mathrm{LH}}$, $p \leq n-1$, $|\psi_i| < \min(\rho_{a_i}, \rho_{b_i})$, $\psi_i \neq 0$, $g \leq B_n < 2g$ for all large n, fixed coefficients. Then, under G with $\sigma^2 > 0$ fixed:*

(a) $1 - \pi_0(S^\circ) = O_{\mathrm{P}_0}(2^B n^{-1/2})$*, hence $\pi_0(\mathcal{R}_d) \to 1$ for every $d \geq 0$;*

(b) $\pi_0(S_0) = p^{-c} n^{-c/2} e^{O_P(1)}$ *(masking), with an exact limit law for the masking statistic (PROOF SKETCH);*

(c) VIS fails at reference S_0, so the favourable-start statement does not apply;

(d) $1 - \pi(S_0) = O_{\mathrm{P}_0}(\sqrt{\log n / n})$ *(locked T03);*

(e) $\pi_0(S_0) \to 0$, $\pi(S_0) \to 1$, $\pi_0(\mathcal{R}_0) \to 1$*: the start is masked, the target is not.*

iid-Gaussian realisation (single unverified run). *With i.i.d. Gaussian rows and $n^a \leq p \leq n^A$ ($p \gg n$ allowed), the partition equals the groups with probability $\to 1$,* $1 - \pi_0(S^\circ) = O_{\mathrm{P}_0}(\sqrt{\log n / n})$ without *the 2^B factor, and the exact limit law of the masking statistic holds (W3-PXN P1–P4, with P5–P8).*

Proof: see Supplement §S.3 (source branch `theory/W1-EXP`, `PROOF.md` §4; `theory/W3-PXN`, `PROOF.md`).

Remark 4 (Discipline). Parts (iii) and (iv)–(v) of Theorem 2 are never combined on one sequence; on LH the reference in (iii) is S°. The cliff $|\psi| < \rho$ (beyond it mass moves to a different regime) and the packing condition $g \leq B < 2g$ are part of the theorem and must be stated. The rates in (iii) carry a factor 2^B and are numerically vacuous below $n \approx 10^9$ (Section 5).

4.4 Theorem 3: the repair (refined T37, T24)

Conditions. For a finite family $\mathcal{R} \ni S_0$ of feasible models: **(L)**: $M(S_0) = \perp$; $\mathrm{TD}'_\infty(\mathcal{R})$: (L) and for every $S \in \mathcal{R} \setminus \{S_0\}$, $M(S) \neq \perp$ and $M(S) \in \mathcal{R}$; $\mathrm{TD}'_P(\mathcal{R})$: (L) and $M(S)$ is a progress move for every $S \in \mathcal{R} \setminus \{S_0\}$. (Strict separation $\bigcap_S D_S$ together with (L) is *sufficient* for TD'_P; with exact ties it is not necessary.) $\mathrm{POS}(\mathcal{R}_d)$: at every $S \in \mathcal{R}_d \setminus \{S_0\}$ the move designated by literal TD(1) or TD(2) has gain > 0.

Lemma 3 (The literal greedy-descent condition is unsatisfiable)**.** *For $d \geq 1$ the literal condition* $\mathrm{TD}(d, m)$ *of the original T37 is unsatisfiable: at $S = S_0 - j + l$ the swap $l \to j$ beats the addition of j whenever $F(S_0) > F(S_0 + l)$, so* $\mathrm{TD}(d, 0)$ *fails for every $d \geq 1$ on $\{\pi(S_0) > \frac{1}{2}\}$, i.e. under the conclusion of Theorem 1. Therefore the literal T37, T37(ii)'s union bound (it includes a comparison with the wrong sign) and "exactly $|S \triangle S_0|$ moves" (swaps cut Φ by 2) are withdrawn.*

Proof: see Supplement §S.4 (source branches `theory/W1-T37R` Prop. 1; `theory/W3-VT38E` V1).

Theorem 3(i) (deterministic greedy repair — Theorem A)**.** *Assume EX-A; X, y fixed; ℓ the production move cap; $\mathcal{R}$ a finite family of feasible models containing S_0.*

(A1) $\mathrm{TD}'_\infty(\mathcal{R})$ *holds if and only if, for every $S \in \mathcal{R}$, the uncapped greedy path from S stays in $\mathcal{R}$ and terminates at S_0.*

(A2) Under $\mathrm{TD}'_P(\mathcal{R}_{a,d})$*, from every $S \in \mathcal{R}_{a,d}$ the greedy path stays in* $\mathrm{Lat}(S) \subseteq \mathcal{R}_{a,d}$*, Φ strictly decreases at each move and S_0 is reached after $\tau(S) = m(S) + e(S) - \sigma(S) \leq \Phi(S)$ moves, where $\sigma(S)$ is the number of extra→truth swaps used;* $\max\{m(S), e(S)\} \leq \tau(S)$.

(A3) If $\mathrm{TD}'_P(\mathcal{R}_{a,d})$ *holds and $\ell \geq a + d$ (or merely $\ell \geq \tau(S_c)$), then from any centre $S_c \in \mathcal{R}_{a,d}$ the production local search returns S_0 with* `local_certified`*, without* `local_limit`*, and afterwards $\widehat{S} = S_0$ up to exact ties (with* no $\pi(S_0) > \frac{1}{2}$ *premise). The same holds for one centre S_c under the lattice condition (L)* $\wedge \bigcap_{S' \in \mathrm{Lat}(S_c) \setminus \{S_0\}} D_{S'}$*, which involves $2^{\Phi(S_c)}$ states only.*

(A5, repaired) Literal strict $\mathrm{TD}(d, 0) \wedge \mathrm{POS}(\mathcal{R}_d) \Rightarrow \mathrm{TD}'_P(\mathcal{R}_d) \Rightarrow \mathrm{TD}'_\infty(\mathcal{R}_d)$*; the converse of the second implication fails.*

Proof: see Supplement §S.4 (source branch `theory/W1-T37R`, `PROOF.md` Theorem A; repairs `theory/W3-VT37R`, `REPAIRS.md` R1, R3).

Lemma 4 (Lemma LS-score)**.** *If D_S holds on* $\mathrm{Lat}(C_0) \cap \{\Phi \geq 2\}$ *and $\ell \geq \Phi(C_0) - 1$, the production search from C_0* scores *S_0 (the next scan at a flip neighbour scores it); certification is not required. This is the form used by Theorem 5.*

Proof: see Supplement §S.4 (source branch `theory/W4-T38FULL`, `T38_PROOF.md` §3.1).

Lemma 5 (Lemma 0, restricted to what survives review)**.** *The final local search, centred at the running HPM, can deliver S_0 (i.e. $S_0 \notin \mathcal{S}_{\mathrm{pre}}$ and $S_0 \in \mathcal{S}$ after the search) only on $E_{\mathrm{LS}} = \{S_0 \notin \mathcal{S}_{\mathrm{pre}},\ F(S_0) > F(\text{centre})\}$, by monotonicity of F along the greedy path. After a certified repair with S_0 the centre, $\widehat{S} = S_0$ up to exact ties.*

Proof: see Supplement §S.4 (source branch `theory/W1-T37R`, `PROOF.md` Lemma 0 (a),(c) only).

Theorem 3(ii) (asymptotic repair — Theorem W)**.** *Assume regime R1; s_0, $a \leq s_0$, d fixed; $\mathcal{D} = \mathcal{D}_n \subseteq S_0^c$ deterministic (X-measurable); EX-A, G, R,* $\mathrm{SE}_\lambda(d+1)$ *($\lambda_{\min}(R_{S_0 \cup J}) \geq \varphi_0^2$ for $|J| \leq d+1$),* BM-∞*, $\boldsymbol{PGD}(\eta)$ on $\mathcal{R}_{a,d}(\mathcal{D}) = \{U \cup P : U \subseteq S_0,\ |S_0 \setminus U| \leq a,\ P \subseteq \mathcal{D},\ |P| \leq d\}$ for a fixed $\eta \in (0, 1]$ (at every state missing a truth, the best non-progress reduction of the population residual is at most $(1 - \eta)$ times the best progress reduction), and one of (H1) $d \leq 1$; (H2) $(|\mathcal{D}| + 1)^{d-1} \leq n^{1/2-\epsilon}$; (H3)* $\mathrm{LD}^\flat(d; B', \rho^*)$ *with $\tau^2 < 1 + 1/(2A)$, $\tau = (1 + \rho^* \sqrt{d})/\sqrt{1 - \rho^{*2}}$. Then* $\mathrm{P}_0\big(\mathrm{TD}'_P(\mathcal{R}_{a,d}(\mathcal{D}))\big) \to 1$*; consequently for every (data-dependent) centre S_c and all n with $\ell_n \geq a + d$,*

$$\mathrm{P}\big(\text{the search from } S_c \text{ does not return } S_0 \text{ certified}\big) \ \leq\ \mathrm{P}\big(S_c \notin \mathcal{R}_{a,d}(\mathcal{D})\big) + o(1).$$

Proof: see Supplement §S.4 (source branch `theory/W3-PGD`, `PROOF.md` §4). The condition LD$^\flat$ is that of `PROOF.md` §3 (bounded neighbour sets $\mathfrak{N}(l)$ of size $\leq B'$ and a squared partial multiple correlation at most ρ^*).

Remark 5 (Limits of Theorem 3(ii)). PGD is not implied by SE + BM-∞ (combination look-alike counterexample; W1-T37R, verified by W3-VT37R), and is sharp: a reverse margin $\rho^0 \geq (1+\epsilon)\rho^+$ on a subsequence forces failure (W3-PGD Prop. P4). The local-dependence condition cannot be dropped for $d \geq 2$ with unrestricted extras (equicorrelated nulls with $r > \frac{1}{2}$: $\mathrm{TD}'_P(\mathcal{R}_{0,2})$ fails with probability $\to 1$; W3-PGD Prop. C-LD, COUNTEREXAMPLE); it is unnecessary for $d = 1$ or for extras from a class of size $n^{o(1)}$.

Theorem 3(iii) (explicit margin on the masked family LH)**.** *On LH (exact Gram, as in Theorem 2(v)) with $S^\circ = S_0 \setminus M$, the population condition* POP-LAT(S°, m_0) *holds with the explicit sharp margin $m_0 = \min\{(1-\kappa_{\max})R_{\min},\ \min_i \beta_i^2(1-\psi_i^2)^2,\ \min_h \theta_h^2\}$, where $R_i = \beta_i^2\psi_i^2(1-\psi_i^2)$ and $\kappa_i = \rho^2(1-\psi_i^2)/(1-\psi_i^2\rho^2)$. Moreover (Lemma M) any population margin $m_n \gg \sqrt{\log n/n}$ turns* POP-LAT *into the event D_S on* $\mathrm{Lat}(S^\circ)$ *with probability $\to 1$; hence the search from S° scores S_0 in c moves.*

Proof: see Supplement §S.4 (source branch `theory/W4-T38FULL`, `T38_PROOF.md` §§3.2, 3.4).

Theorem 3(iv) (the Gibbs path — Theorem B$'$ / B$^\sharp$)**.** *Assume EX-A, EX-S, SR, $\phi = 1$. For a particle at a state $x \in \Omega \setminus \{S_0\}$ at the start of H terminal rounds at $\phi = 1$ without resampling in between, with the exact data functions $c_*(x)$, $v(x)$ and $\eta(x)$ of the Supplement ($v(x)$ is a two-phase potential, $\leq |M(x)| + a$ while a truth is missing, and $|P|$ on supersets):*

$$\mathrm{P}_x(S_0 \text{ not scored within } H \text{ rounds}) \ \leq\ \mathrm{P}\big(\mathrm{Bin}(H, c_*(x)) < v(x)\big) + H\eta(x) \ \leq\ v(1-c_*)^{\lfloor H/v \rfloor} + H\eta,$$

with $c_2 \geq 1/(B+3) - \varepsilon_{\mathrm{del}}$ deterministically on supersets, and $c_1 \geq 1/|M(x)|$-type bounds from pointwise addability (no factorial). Terminal-round trajectories of different particles are conditionally independent given the terminal-start population and past (IND-T, under EX-S), so the population failure probability is the product of the per-particle bounds for particles selected using the past only (Cor. B-M). At Hamming distance one the truth is scored in the next round with probability at least $a_1(x)$ (Cor. P1).

Proof: see Supplement §S.5 (source branches `theory/W1-T24G` §4, `theory/W2-VT24G` Lemma F$'$, B1$'$/B2$'$, `theory/W2-PREM` IND-T). *Status caveat.* The exact definitions of $c_*(x)$, $v(x)$, $\eta(x)$ are those of W1-T24G §4 as repaired; W6-FIXREV1 (which corrects Proposition U's classification or charging) had not delivered its package when this draft was finalised; the statement of v above is therefore the source's inequality form, and the exact constants must be re-read from the W6-FIXREV1 package before submission.

Remark 6 (Asymptotic premises of Theorem 3(iv)). The asymptotic status of the inputs to Theorem 3(iv) is mixed and must not be summarised as "$H\eta \to 0$". τ_2, NL0(0), UDEL(1): PROVED. NL0(a)/UDEL($a \geq 2$): PROVED *under* a partner-graph condition PG and FALSE without it (equicorrelated nulls). TS: PROVED under non-cancellation, FALSE under a cancellation pair. IND-M: PROVED under subset-start concentration and a non-degeneracy condition (SCR + GS), OPEN for superset starts. On LH all premises are PROOF SKETCH (two population inequalities only numerical). The older uniform exit hypothesis H24 is *retired*: at boundary states of $\mathcal{R}_d$ a look-alike of a missing truth exits with probability $\geq (1-o(1))/(B+d+4)$ (W1-T24G CE1, verified as a deterministic bound; the broad asymptotic wording for correlated designs is a GAP per A-REV1, to be replaced by a narrow statement), and the form $(1-c')^{\lfloor H/(s_0+d)\rfloor}$ of the brief is invalid (CE2).

4.5 Theorem 4: HPM composition (T21)

Lemma 6 (Run/output split)**.** *Under GUARD, for an output-available event* $\mathcal{O} = \mathcal{G}^c \cap \bar{\mathcal{O}}$ *with* $\bar{\mathcal{O}} \in \sigma(\bar{R})$*, a reported output* g *and its potential version* $\bar{g}$*:* $\mathbf{1}_{\mathcal{O}} g = \mathbf{1}_{\mathcal{O}} \bar{g}$*;* $\mathrm{P}(\mathcal{O}^c \cup \{g \notin A\}) \le \mathrm{P}(\mathcal{G}) + \mathrm{P}(\bar{\mathcal{O}}^c) + \mathrm{P}(\bar{\mathcal{O}}, \bar{g} \notin A)$*; and* $\mathrm{E}[\mathbf{1}_{\mathcal{O}} g] \le \mathrm{E}[\mathbf{1}_{\bar{\mathcal{O}}} \bar{g}]$ *for* $g \ge 0$*. Identities in expectation are not preserved under* $\mathrm{P}(\cdot \mid \mathcal{O})$*.*

Proof: see Supplement §S.6 (source branch `theory/W2-BOOK`, `PROOF.md` §0.2).

Theorem 4 (Composition for the reported HPM, T21)**.** *Assume EX-A, EX-S, CAL, SR and GUARD. Let* $U \supseteq \mathcal{G}$ *be the no-output event,* $\bar{U}$ *its potential-run part,* $\bar{\mathcal{S}}$ *the potential scored set and* $\delta_1 = 1 - \pi(S_0)$*. Then, for* every n *and every data set,*

$$\mathrm{P}(U \text{ or } \widehat{S} \ne S_0) \;\le\; \mathrm{P}(U) + 2\,\mathrm{E}_0\delta_1 + \mathrm{P}(\bar{U}^c,\; S_0 \notin \bar{\mathcal{S}}), \qquad \mathrm{P}(U) \le \mathrm{P}(\mathcal{G}) + \mathrm{P}(\bar{U}),$$

equivalently $\mathrm{P}(\mathcal{O} \cap \{\widehat{S} \ne S_0\}) \le 2\mathrm{E}_0\delta_1 + \mathrm{P}(\bar{\mathcal{O}},\; S_0 \notin \bar{\mathcal{S}})$ *(Lemma 6). The last term,* $\mathrm{P}(\bar{U}^c, S_0 \notin \bar{\mathcal{S}}) = \mathrm{P}(\bar{\mathcal{O}}, S_0 \notin \bar{\mathcal{S}})$*, is bounded by each of:*

(a) (i.i.d. final draws) $\mathrm{P}(\mathcal{A}^c, S_0 \notin \mathcal{S}_{\mathrm{cal}}) + \mathrm{E}\big[\mathbf{1}_{\mathcal{A}} \mathbf{1}\{S_0 \notin \mathcal{S}_{\mathrm{cal}}\}(1 - \pi_0(S_0))^{Q_f}\big]$*, with* $Q_f = 4N_e(T + L + 5)$*;*

(d) (tables and scans) the probability that S_0 *lies in no block table and no local search of* $\bar{R}$ *is centred at a flip or swap neighbour of* S_0*;*

(d^+*) (with Theorem 3(i)) if a data event* TD^* *is such that, on* TD^**, every completed local search of* $\bar{R}$ *centred in* $\mathcal{R}_d$ *with* $\ell \ge |c \triangle S_0|$ *scores* S_0*, then* $\mathrm{P}(\bar{\mathcal{O}}, S_0 \notin \bar{\mathcal{S}}) \le \mathrm{P}_0(\mathrm{TD}^{*c}) + \mathrm{P}(E_{\mathrm{LS}}^c \cap \mathrm{TD}^*) + \mathbf{1}\{\ell < s_0 + d\}$*, where* E_{LS} *is the event that some local search of* $\bar{R}$ *has its centre in* $\mathcal{R}_d$ *(drop the last term under the extra-scan rule). With LS0 the start-side event* $\{\tilde{S}_0 \in \mathcal{R}_d\} \subseteq E_{\mathrm{LS}}$*;*

(e'*) (Gibbs-path repair at a past-measurable selected state* x*)* $\mathrm{P}(\mathcal{A}^c, S_0 \notin \mathcal{S}_{\mathrm{cal}}) + \mathrm{P}(\mathcal{A}, \text{no eligible particle}) + \mathrm{E}[\mathbf{1}_{A_F} \min\{1, \beta_H(X_{\mathrm{sel}})\}]$*, with* $\beta_H(x) = \mathrm{P}(\mathrm{Bin}(H, c_*(x)) < v(x)) + H\eta(x)$ *from Theorem 3(iv) and* $H \le L + 5$*; and, for eligible particles* i *at the boundary,* $\prod_i \min\{1, \beta_H(X_i)\}$ *in place of* $\min\{1, \beta_H(X_{\mathrm{sel}})\}$ *(IND-T).*

The earlier route (e) with H24 is retired. The premise $\pi(S_0) > \frac{1}{2}$ *is not needed when the local search certifies* S_0 *(Theorem 3(i)(A3)).*

Proof: see Supplement §S.6 (source branches `theory/W2-BOOK`, `PROOF.md` §4 and `STATEMENTS.tex`; `theory/W2-VT24G` §6; locked T21).

Remark 7 (With LS0). The same inequality holds with $\widehat{S}^+$, $\mathcal{S}^+$ and $\bar{\mathcal{S}}^+$ in place of $\widehat{S}$, $\mathcal{S}$, $\bar{\mathcal{S}}$ (the strict-maximiser argument is unchanged): $\mathrm{P}(\mathcal{O} \cap \{\widehat{S}^+ \ne S_0\}) \le 2\mathrm{E}_0\delta_1 + \mathrm{P}(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+\})$ (Theorem 5(0)).

Remark 8 (Reporting $\mathrm{P}(U)$ and the no-conditioning rule). $\mathrm{P}(U)$ must be reported empirically: $6/2{,}400 = 0.25\%$ (Wilson 95% interval $[0.11\%, 0.54\%]$); all six stops are scenario S5; S5 at $p = 1000$ shows 2.5%. The stops are counted as misses. Never condition on completion: a 0.22% stop rate can make $\mathrm{E}[\widehat{Z} \mid \text{output}] = 0.974\,Z$ (W2-BOOK exact counterexample).

4.6 Theorem 5: consistency of the reported HPM (T38) — the main theorem

Hypotheses, in groups. *Operational:* EX-A, EX-S, CAL, SR, GUARD (Section 2); LS0 (Definition 2) is part of the method. *Posterior:* **(P)** $\delta_1 = 1 - \pi(S_0) \to 0$ in P_0-probability, on a triangular array R1 (sufficient: the hypotheses of Theorem 1). *Favourable regime:* **(F)** $1 - \pi_0(S_0) \to 0$ in probability (sufficient: (F1), (F2) or (F3) of Theorem 2(iii) at reference S_0). *Masked regime, route (b1):* there is a *deterministic* reference $S^\circ \in \mathcal{R}_d$ with $\Phi(S^\circ)$ bounded such that

(M1) **ELIG**(S°): $\mathrm{P}_0(\pi_0(S^\circ) > \frac{1}{2}) \to 1$ (sufficient: G, R-lite, SE-blk, ZL(S°), SPREAD, VIS-lin(S°), by Theorem 2(iii)(F3) at S°);

(M2) POP-LAT(S°, m_n): for every $S \in \mathrm{Lat}(S^\circ)$ with $\Phi(S) \geq 2$ some progress neighbour A_S of S satisfies $\Delta(B) - \Delta(A_S) \geq m_n$ for every non-progress neighbour B of S and for $B = S$, with $m_n \geq M_n\sqrt{\log n/n}$, $M_n \to \infty$;

(M3) $\lambda_{\min}(R_{S^\circ \cup S_0}) \geq \varphi_0^2 > 0$ and $\|\mu\|^2/n \leq M$;

(OP-LS0) LS0 is executed, with its centre emitted, whenever output is available.

Theorem 5 (Two-regime consistency of the reported HPM, T38)**.** *Assume EX-A, EX-S, CAL, SR, GUARD and (P) on a triangular array R1 with the v3 budgets indexed by n.*

(0) ***Composition, every*** *n.* $\mathrm{P}(\mathcal{O} \cap \{\widehat{S}^+ \neq S_0\}) \leq 2\mathrm{E}_0[\delta_1] + \mathrm{P}(\bar{\mathcal{O}} \cap \{S_0 \notin \bar{\mathcal{S}}^+\})$. *Hence, whenever the discovery term tends to* 0, $\mathrm{P}(\mathcal{O} \cap \{\widehat{S}^+ \neq S_0\}) \to 0$ *and therefore* $\liminf_n \mathrm{P}(\mathcal{O} \cap \{\widehat{S}^+ = S_0\}) \geq 1 - \limsup_n \mathrm{P}(U)$.

(a) ***Favourable regime.*** *Under (F) the discovery term tends to* 0*, with or without LS0 (the final run's $Q_f \geq 4N_e(L+6)$ i.i.d. π_0 draws score S_0; T11(iii)).*

(b1) ***Masked regime*** *($\pi_0(S_0) \to 0$ allowed, at the rate of Theorem 2(iv)),* ***route (b1), LS0.*** *Under (M1), (M2), (M3), (OP-LS0) and G, on the corresponding events $\tilde{S}_0 = S^\circ$ and LS0 scores S_0 after at most $\Phi(S^\circ) - 1$ moves; the cap $\ell_n \geq \Phi(S^\circ) - 1$ suffices. Hence the discovery term tends to* 0.

(b1-LH) ***An explicit family on which every premise of (b1) holds jointly.*** *On LH in its exact-Gram realisation (Theorem 2(v): $p \leq n-1$, $|\psi_i| < \min(\rho_{a_i}, \rho_{b_i})$, $\psi_i \neq 0$, $g \leq B_n < 2g$, fixed nonzero coefficients), with $S^\circ = S_0 \setminus M$: (P), (M1), (M2) with a fixed margin $m_0 > 0$ (Proposition 3(iii)) and (M3) all hold; $\pi_0(S_0) = p^{-c}n^{-c/2}e^{O_P(1)}$; and $Q_f\pi_0(S_0) \to 0$ when $c \geq 2$ and $p \to \infty$, and also for $c = 1$ when $p/(\sqrt{n}\log^2 n) \to \infty$, so that route (a) fails. With LS0,* $\mathrm{P}(\mathcal{O} \cap \{\widehat{S}^+ \neq S_0\}) \to 0$ *and no statistical input is left open.*

(c) ***Cost.*** *The number of score evaluations is bounded deterministically by T32(i)'s ceiling plus $(\ell+1)\max_{k \leq K}[p + k(p-k)]$ for LS0; under the v3 budgets this is $O((n+p)(p+B)\log^2 n + p^2\log^2 n)$, polynomial in n under R1.* Conditional on T32(i), which is a CONJECTURE.

Not claimed: *evidence or PIP accuracy in the masked regime; anything given completion, or for fits selected on diagnostics; comparative superiority; any finite-n error rate at $n = 200$; a masked $p \gg n$ instance until the single-run LH-iid argument is verified (Theorem 2(v), iid realisation, is a* candidate input*, not part of this theorem); masked-regime consistency* without *LS0 on any explicit family (OPEN).*

Proof: see Supplement §S.7 (source branch `theory/W4-T38FULL`, `T38_PROOF.md` §§1–3, 6–7; inputs: Theorems 4, 2(iii), 3(i), Lemma 4, Proposition 3(iii)).

Remark 9 (Conditional routes — not part of Theorem 5). Two further routes would remove the discovery term with *no* method change, but both are conditional and A-REV1 classifies the supplied formulation as a GAP (to be repaired by W6-FIXREV1). **(b1′) Pilot first stage (D2).** If a one-flip neighbour $x^\circ = S_0 \oplus j$ has $\pi_0(x^\circ) \geq b > 1/10$, the dependence parameter satisfies $\mathrm{Dep} \leq 1 - c_1$ off a data-measurable set $\mathcal{D}$ with $|\mathcal{D}| \leq k_D$, and the no-blow-up event NB′ (effective sample size of the initial pilot weights $\geq 0.9H$, or S_0 drawn) has probability $\to 1$ (with the corrected

constant $r^* = (11 + 2\sqrt{10})/9$), then the first stage of the tier-0 pilot scores S_0: $\mathrm{P}(S_0 \notin \mathcal{S}_{\mathrm{cal}}) \le e^{-2Ht^2} + \mathrm{P}(\mathrm{NB}'^c) + \exp\{-H(1-\mathrm{Dep})_+ g(b-t)/(k_D+B+3)\}$. A statistical instance exists only as a candidate on LH with $c = 1$ (D2-a,b,c PROVED by W3-PXN, unverified; NB: PROOF SKETCH); $c \ge 2$ is OPEN. **(b2) Gibbs path.** With a data-measurable eligible set and checkable envelope hypotheses (potentials with $(K_t h_t) \ge (1-\eta_t)h_{t-1}$, confinement sets with $\mathrm{P}(E^c) \le \varepsilon_E$, fitness oscillation $\Gamma + \sum_t \eta_t \le \log(N\pi_0(h_0)/T_e) - \omega(1)$) and (REP) $\sup_{x\in W} \beta_{L+5}(x) \to 0$, then $\mathrm{P}(S_0 \notin \mathcal{S} \mid \mathcal{C}) \le [\varepsilon_E + \exp\{-N\pi_0(h_0)(1-e^{-c})\}]^4 + \varepsilon_{\mathrm{rep}}$ (W4-SHARE, PROVED, unverified; no explicit family has these inputs verified: OPEN). The earlier "bridge mass ⇒ delivery" route is false (C1, COUNTEREXAMPLE, verified by W3-VT38E with a reversible kernel).

4.7 Theorem 6: evidence and PIP consistency in the favourable regime (T26(i)–(ii))

Theorem 6 (Evidence and PIP consistency, T26(i)–(ii))**.** *Assume EX-A, EX-S, CAL, START, OUT (Remark 1), and*

(Ω) $\mathrm{P}_0(s_0 \le K_n) \to 1$*;*

(P1) for every $\varepsilon > 0$, $\mathrm{P}\big(\mathcal{G}_n \cap \{(T_n+1)\Delta_n > \varepsilon\}\big) \to 0$ *(joint probability over data and calibration streams;* T_n *the* realised *stage count; sufficient deterministic-budget form* $(T_n^{\max}+1)\Delta_n \to 0$ *in* P_0*-probability), with* $\Delta_n = \max\{1-\pi_0(S_0), 1-\pi(S_0)\}$ *(Theorem 2(i));*

(P1-ex) $\mathrm{P}_0(\mathcal{E}_n \cap \{\delta_1 > \varepsilon\}) \to 0$ *for every* $\varepsilon > 0$ *(the exact-branch twin of (P1)).*

Here $\mathcal{G}_n$ *is the event that the SMC branch is taken and the calibration succeeds,* $\mathcal{E}_n$ *the exact-enumeration branch. No statistical assumption is used by this theorem. Then,* never conditioning on completion*:*

(A-0) (fixed n*, calibration-conditional) for every island* r*:* $\mathrm{E}_{\mathcal{C}}|\widehat{Z}_r^{\circ}/Z - 1| \le b(\mathcal{C}) := 2(T+1)\Delta_n/(1-\Delta_n)$*; the occupancy fraction off* S_0 *at the end of stage* T *and at the start of each measured round has* $\mathrm{E}_{\mathcal{C}} f \le \min\{1, 2\delta_1 + 2b(\mathcal{C})\}$*;* $\mathrm{E}_{\mathcal{C}} \max_j |\widehat{\mathrm{PIP}}_{rj}^{\circ} - \mathbf{1}\{j \in S_0\}| \le \min\{1, 2\delta_1 + 2b(\mathcal{C})\} + \bar{\rho} + u_\pi$ *with* $\bar{\rho} \le (B+3)u_\pi$ *and* $u_\pi = \delta_1/\pi(S_0)$*;* $\mathrm{P}_{\mathcal{C}}(\widehat{\mathrm{MPM}}^{\circ} \ne S_0) \le 2[\min\{1, 2\delta_1 + 2b(\mathcal{C})\} + \bar{\rho} + u_\pi]$*;*

(A-i) for every $\varepsilon > 0$, $\mathrm{P}(\mathcal{O}_n \cap \{|\widehat{Z}/Z - 1| > \varepsilon\}) \to 0$ *(also islandwise and for* $\log \widehat{Z}$*);*

(A-ii) for every $\varepsilon > 0$, $\mathrm{P}(\mathcal{O}_n \cap \{\max_j |\widehat{\mathrm{PIP}}_j - \mathbf{1}\{j \in S_0\}| > \varepsilon\}) \to 0$ *and* $\mathrm{P}(\mathcal{O}_n \cap \{\widehat{\mathrm{MPM}} \ne S_0\}) \to 0$*.*

The frozen second premise $(L_n+4)\rho_n^* \to 0$ *is redundant.* ***Not claimed:*** $\mathrm{P}(\mathcal{O}_n) \to 1$*; any statement in the masked regime; the secondary* ($\widehat{Z}_r$*-weighted) PIP; the HPM (Theorem 5 replaces the old T26(iii)); uniform Monte Carlo rates in* N*; anything about runs selected by their flags or completion; floating-point validity.*

Proof: see Supplement §S.8 (source branch `theory/W2-T26`, `PROOF.md` §§1–3).

Remark 10 (Where the premise comes from, and its cost). In R1 under the hypotheses of Theorem 1 and (F1) of Theorem 2(iii), (P1) holds in its deterministic-budget form, $(T_n^{\max}+1)\Delta_n = O_P\big(\log^2 n\,[2^B\sqrt{\log n}\,n^{-\kappa} + e_n + \sqrt{\log n/n}]\big) \to 0$ (with $e_n = o_P(n^{-m})$ for all m); the ORTH family ($X^\top X = (n-1)I$, $p \le n-1$, fixed θ) satisfies all premises jointly (PROOF SKETCH). Premise (P1) must be stated with the *realised* T_n (random), not with $T^{\max} = 4{,}000$: with the deterministic cap the quantity is vacuous below $n \sim 10^8$–10^9.

4.8 Proposition 2: frozen-run identities and the reported bound (T12, T31(i))

Proposition 2 (Frozen-run identities and the reported bound)**.** *Assume EX-A, EX-S, START and CAL, with the calibration $\mathcal{C}$ (tier, grid, T, N_e, L, round and island counts, local-search cap, inherited score leaders and every other setting read by the final run) a function of the data and calibration streams only, and escalation decided before any final randomness. For the potential run, on the success event $\mathcal{A}$ and for every island r and variable j:*

(a) $\mathrm{E}[\widehat{Z}_r \mid \mathcal{C}] = Z$ *and* $\mathrm{E}[\widehat{Z}_r \widehat{\mathrm{PIP}}_{rj} \mid \mathcal{C}] = Z\,\mathrm{PIP}_j$*;*

(b) $\mathrm{E}[\widehat{Z} \mid \mathcal{C}] = Z$ *and* $\mathrm{E}[\frac{1}{4}\sum_r \widehat{Z}_r \widehat{\mathrm{PIP}}_{rj} \mid \mathcal{C}] = Z\,\mathrm{PIP}_j$*;*

(c) $\mathrm{E}[\widehat{Z}\,\bar{P}_j \mid \mathcal{C}] = Z\,\mathrm{PIP}_j - \frac{3}{4}\mathrm{Cov}(\widehat{\mathrm{PIP}}_{1j}, \widehat{Z}_1 \mid \mathcal{C})$ *and* $\mathrm{E}[\bar{P}_j \mid \mathcal{C}] - \mathrm{PIP}_j = -\mathrm{Cov}(\widehat{\mathrm{PIP}}_{1j}, \widehat{Z}_1 \mid \mathcal{C})/Z$*, where $\bar{P}_j = \frac{1}{4}\sum_r \widehat{\mathrm{PIP}}_{rj}$ is the reported PIP; both are non-zero in general (so "pooled evidence $\times$ pooled PIP" is* not *an identity);*

(d) $\mathrm{P}(Z \geq \alpha\widehat{Z} \mid \mathcal{C}) \geq 1 - \alpha$ *for $0 < \alpha < 1$ (also islandwise), and* $\mathrm{E}[\log \widehat{Z} \mid \mathcal{C}] \leq \log Z$*.*

Under GUARD, for the reported evidence, $\mathrm{P}(\mathcal{O}_E,\ \alpha\widehat{Z} > Z) \leq \alpha\mathrm{P}(\mathcal{A})$ *and* $\mathrm{E}[\mathbf{1}_{\mathcal{O}_E}\widehat{Z}] \leq Z\mathrm{P}(\mathcal{A})$*. If moreover the floating-point evidence satisfies $|\log \widehat{Z}^{\mathrm{fp}} - \log \widehat{Z}| \leq \eta$ outside an event of probability ε within $\mathcal{O}_E$ (the validation contract VAL-FP(η, ε)), then* $\mathrm{P}(\mathcal{O}_E,\ \alpha\widehat{Z}^{\mathrm{fp}} > Z) \leq \alpha e^{\eta} + \varepsilon$*. The secondary ($\widehat{Z}_r$-weighted) PIP is removed.*

Proof: see Supplement §S.9 (source branch `theory/W2-BOOK`, `PROOF.md` §1; locked T12, T31(i)).

Remark 11 (What is not claimed)*.* No unbiasedness given completion is claimed: $\mathrm{E}[\widehat{Z}^{\mathrm{rep}} \mid \mathcal{O}_E] = Z$ is false in general (W2-BOOK exact counterexample: 0.22% stops give $\mathrm{E}[\widehat{Z} \mid \text{output}] = 0.974\,Z$). The reported PIP is not unbiased for PIP_j (part (c)). The VAL-FP contract has not been measured.

5 What the theory does and does not say at $n = 200$

The theorems are statements about mechanism and about $n \to \infty$; they are not statements about the reported numbers at the study's size ($n = 200$, $p \in \{500, 1000\}$, $s_0 = 14$). The simulations are evidence in their own right. The following facts must be read with every theorem above.

5.1 What the theory does say

The only non-vacuous *finite-n* statements are exact inequalities: the composition (C1) of Theorem 4 (including its form with LS0, Remark 7), the deterministic Theorem 3(i) (with the lattice form of (A3), and Lemma 4), the ESS-share inequality of the pilot analysis, Corollary P1 of Theorem 3(iv) at $B = 4$, and Proposition 2. In particular, the decomposition (1) is exact, and at the study's size the error is carried by its first term (item 2 of the list in Section 5).

5.2 What the theory does not say

1. **The start at $n = 200$ is dilution-masked, not cancellation-masked.** The block tables score against the full response, so each truth sees the other truths as noise ($v_T \approx 7$–$26\sigma^2$); the start mode misses 6–12 of the 14 truths at every signal scale, and this disappears only for n of order 800–1600 in these designs. The masked start of Theorems 2(iv) and 2(v) is a different mechanism (cancellation across blocks), and LH's cancellation cannot survive the study's

partition (groups of 5 with $B = 14$: co-blocking kills LH masking, W3-PXN Prop. Q, PROOF SKETCH; LH needs $g \leq B < 2g$). The paper must therefore present Theorems 2(iv)–2(v) and 5(b1) as the masked regime the sampler is designed for, *not* as the study's masking.

2. **Every completed wrong HPM is a posterior miss.** In the $p \gg n$ study, 565 of 2,394 completed fits have $F(S_0) \leq F(\widehat{S})$ (0 discovery misses); there are 6 resource stops, all in S5, counted as misses ($6/2{,}400 = 0.25\%$). So the binding term at this size is the posterior term $2\mathrm{E}_0\delta_1$ of (1); the repair statements (Theorems 3(i), 3(ii)) must be read "whenever the posterior allows".

3. **Theorem 1 is asymptotic, with proof thresholds far above $n = 200$.** The deterministic steps of the proof, as written, hold only from $n \approx 2.1 \times 10^6$ ($p = n^{1.17}$) to 2.5×10^6 ($p = n^{1.30}$): the binding condition is $q_n \leq \frac{1}{2}$ (the earlier figure 1.2×10^5 is an ERROR, corrected by W2-VT03R and confirmed by A-REV2); the O_P constants are unspecified. The sparse-eigenvalue condition SE_w ranges over at most 114 columns at $n = 200$ *because the cap* $K = 100$ *binds*, not because of the exact size tail (uncapped $m^\sharp = 146/168$); it is unverified on the realised designs (greedy-adversarial eigenvalue probes: $\lambda_{\min}(G) \leq 0.40$ for S1, ≤ 0.22 for S4 at $|S_0 \cup J| = 114$; feasibility of any fixed g_0 is unverified). By interlacing, it requires every 100 columns of X to be linearly independent. The beta-min constant of the proof exceeds the executed signal by at least 43× (proof needs $\theta^{*2} \geq 27.5/\varphi_0^2$ at $p = 500$; executed $\theta^{*2} = 0.64/0.16$), and the orthogonal-design scale $(\log n + 2\log p)/n \approx 0.09$ is 7× *below* the executed signal, so the shortfall is the constant, not the signal.

4. **The start rates are vacuous at the study size.** The rates of Theorem 2(iii) carry a factor 2^B: $2^B\sqrt{\log n/n}$ at $B = 14$ equals 2.7×10^3 at $n = 200$, 497 at $n = 10^4$, 61 at $n = 10^6$, so they are numerically vacuous below $n \approx 10^9$. (Only the 2^B-free iid-Gaussian rate of Theorem 2(v), a single unverified run, avoids this.) Theorem 2(iv) bounds i.i.d. draws from π_0 only, and says nothing about the hybrid sampler.

5. **The HPM at this size is found by Gibbs flips, a route with no completed theorem.** 98% of fresh scores in the study come from Gibbs flips; the final local search repaired 0 fits; whether the study's 541 final centres lay in E_{LS} is unextracted (OPEN). The paper proves exact finite-n inequalities for this route (Theorem 3(iv)) but no rate: the single-particle bound is vacuous at $n = 200$ ($H\eta \approx 4$–16; $v > H$ possible). The route (b2) of Remark 9, which would match the observed behaviour, has no verified instance. The local search is not the delivery mechanism at $n = 200$, and "the local search repairs" must not appear as an explanation of the numbers.

6. **LS0 does not change the picture at $n = 200$.** LS0 cannot fix posterior misses; the start mode is 8–14 moves from S_0 at $n = 200$ (not a bounded $\Phi(S^\circ)$), the cap $\ell = 10$–11 binds in S4/S6, and post hoc on regenerated study data LS0 scored S_0 in 113/128 fits with $|\tilde{S}_0 \triangle S_0| \leq \ell + 1$ and 13/32 beyond. The approval of LS0 is conditional on the O-LS0 data check (few or no changed reported HPMs, never turning a correct report into an incorrect one beyond posterior-mode corrections, negligible compute), whose result was not available at drafting. LS0 buys the one masked-regime consistency statement whose premises are all proved on an explicit family (Theorem 5(b1-LH)); it buys nothing for route (b2) or for evidence/PIP claims in the masked regime.

7. **Evidence and PIP consistency are favourable-regime, long-hypothesis statements.** At $n = 200$, $\Delta_n \approx 1$ (dilution) so Theorem 6 is silent, although the sampler still estimates Z

within 1–2% in the masked design. (P1) must be stated with the realised T_n.

8. **Statements the paper must not make.** (i) The literal T37/TD(d, m), "exactly $|S \triangle S_0|$ moves", the 2^{s_0} union bound, "T37 replaces GAIN, LK, EXIT" (true only for the local-search route and only on E_{LS}); (ii) H24 or any uniform-over-$\mathcal{R}_d$ exit hypothesis, the $(1-c')^{\lfloor H/(s_0+d) \rfloor}$ form, T24's factorial; (iii) "the exact BB tail makes SE meaningful at $n = 200$" (the cap does, and only for SE_w; anchoring is cosmetic, Remark 2); (iv) any finite-n error rate at $n = 200$ from Theorems 1, 2, 3(ii), 5 or 6; (v) "the correlated-design advantage follows from T37's margin" (contradicted at $n = 200$ by the $c = 0.5$ posterior misses in S4–S6; the LH margin of Proposition 3(iii) is the mechanism on LH, with no statement about the study); (vi) "T27 is a lower bound for the method" (it bounds i.i.d. start draws only); (vii) "T26 with budgets indexed by n" via $T^{\max} = 4{,}000$; (viii) $\mathrm{E}[\widehat{Z} \mid \text{completed}] = Z$ and "pooled $\widehat{Z}\times$ pooled PIP is unbiased" (both false); (ix) the cost corollary as a theorem (T32(i) is a CONJECTURE); (x) the word "VERIFIED" as if it meant reviewed: all verification by the campaign is same-family (Claude checking Claude) except the two cross-family reviews A-REV1 and A-REV2, which cover only the items whose verdicts are quoted above; (xi) "D2 covers one-cancellation masking at $p \gg \sqrt{n}$" (conditional; its CF instances are not masked); (xii) "no method change needed": true for everything except Theorem 5(b1), which is the headline and rests on LS0.

9. **Implementation facts the theorems lean on have had at most one reading.** SR (HQ's read of `step_gibbs`: current and flipped configurations are evaluated before the accept draw; no call site disables emission), GUARD (stop-only; code reading of `v3_runtime.h`, `v3_engine.h`, `v3_core.cpp`), the scan-before-limit rule of the local search, IND-T's sequential stream reading, and LS0's `emit_centre`. None has had an independent audit confirmed on these branches.